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An analysis of electronic dephasing in the spin-boson model

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In order to develop a more complete understanding of the limitations of mixed quantum-classical simulation methods, the origins of electronic dephasing are analyzed in a simple model of the condensed phase, namely, the spin-boson model with an ohmic spectral density. We focus on the decay of the thermally averaged nuclear overlap/phase function (NOPF). Considering the strong coupling/high temperature limit, a relationship is obtained at short time between the rate of electronic coherence loss and the electronic dephasing rate characteristic of a classical bath. Using this relationship, we clarify the origin of the decay of the NOPF. In the same limit, we also reproduce an earlier relationship between the electronic decoherence time and a solvation relaxation time. Finally, we point out that, for the spin-boson model, the exact quantum mechanical description of electronic dephasing is reproduced by mixed quantum/classical methods if a Gaussian distribution of quantum fluctuations around each classical phase space point is introduced. That spatial distribution of quantum fluctuations is functionally the same as that appearing in the Feynman–Kleinert variational local harmonic approximation, and also that implemented in existing classical trajectory-based estimates of coherence dissipation times. © *2004 American Institute of Physics.* [DOI: 10.1063/1.1742979]

I. INTRODUCTION

In full quantum mechanical (FQM) dynamics, nuclear bath evolution on alternative electronic potential energy surfaces (PES) leads, in general, to electronic coherence loss, i.e., the decay of off-diagonal elements of the electronic reduced density matrix.[1–3] This so-called electronic dephasing, or decoherence, is comprised of contributions that can be, at least conceptually, separated. In standard mixed quantum/classical (MQC) dynamics simulation, where the bath phase space distribution and dynamics are both classical, there are contributions that arise from the corresponding (classical bath) distribution of electronic energy gaps. These contributions are included in the decay process sometimes referred to as pure dephasing (free of contributions from electronic population dynamics). There are, in addition, contributions to coherence loss that arise from quantum bath effects, which require a description of nuclear bath wave functions.[4] Further, for a thermal system, any contribution to electronic dephasing that is associated with the existence of a thermal ensemble of electronic energy gaps will necessarily increase with increasing temperature, due to increased fluctuations, while effects associated with quantum aspects of the bath must weaken with rising temperature.[5] We note that we have reserved the term ''decoherence'' for these latter, quantum bath, effects in some earlier studies, but this nonstandard distinction is not retained in the present paper.

Because of the expediency of MQC simulation for complex many-body systems, there is substantial motivation to reveal the separations described above in detail. Approaches exist for the implementation of dissipation of electronic coherence in an MQC molecular dynamics simulation,[4,6–8] but either an *a priori* knowledge of the characteristic time scale or a means for calculating that time from classical bath information is required. Most importantly, the contributions that are present in MQC dynamics must be separated from those that are not, in order to correctly incorporate the missing dissipation.[4,8] The purpose of the present paper is to carry out an analysis in detail for a simple model for which such analysis can largely be done analytically. For our present purposes, we consider a two level electronic system linearly coupled to a harmonic bath, the so-called spin-boson model. This model has been used extensively to study dissipation due to system-bath coupling.[9] We also use an ohmic spectral density for the bath, allowing significant analytical development.

Qualitative discussions along these lines have appeared,[10] but few studies have been quantitative.[11] For a thermally equilibrated initial state, MQC methods use the classical Boltzmann weights to define an initial configuration. Alternatively, one can employ a quantum statistical representation for the initial distribution, such as the canonical Wigner distribution.[12] Lockwood *et al.*[10] have considered the spin-boson model in the strong coupling case[13] (large nuclear displacement between levels, corresponding also to a short time, or high temperature, approximation). They have shown that combining the canonical Wigner distribution function for a harmonic oscillator bath with the classical dynamics form of the thermally averaged nuclear overlap/phase function (NOPF) (see below), one obtains an expression equivalent to that obtained at the same level of approximation using the classical Boltzmann weights and a wave packet descrip-

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tion of nuclear initial conditions. We expand on that result here.

The next section considers alternative expressions for the thermally averaged NOPF based on the spin-boson model. In Sec. III, contributions to electronic decoherence times are obtained for the strong coupling case, and the relationship among those are discussed. A relationship between the decoherence time and a solvent relaxation time is obtained in the same limit and shown to correspond to an earlier expression due to Prezhdo and Rossky.[3] Finally, Sec. IV develops the electronic dissipation resulting from the use of a Wigner distribution within an MQC method. We present a summary and conclusions of our study in the final Sec. V.

II. GENERAL CONSIDERATION OF THE CANONICAL NUCLEAR OVERLAP/PHASE FUNCTION

We begin by considering two electronic surfaces which are harmonic in the nuclear coordinates and differ only by displacement of the equilibrium nuclear positions. The two electronic states are the states of the quantum subsystem of particular interest and the nuclei compose a harmonic bath. Much literature has been devoted to consideration of this spin-boson model[9] or displaced harmonic oscillator model,[14] due to the fact that analytical expressions for physical quantities can be obtained. The Hamiltonian of the system is

$$\hat{H}=|i\rangle\hat{H}_i\langle i|+|f\rangle\hat{H}_f\langle f|+\hat{\gamma}^{\dagger}|i\rangle\langle f|+\hat{\gamma}|f\rangle\langle i|, \tag{1}$$

where $|i\rangle$ denotes the initial electronic state and $|f\rangle$ the final electronic state. The electronic coupling matrix element is denoted by $\hat{\gamma}$, while $\hat{H}_i$ and $\hat{H}_f$ are the nuclear Hamiltonians on electronic surfaces $|i\rangle$ and $|f\rangle$, respectively,

$$\hat{H}_i=\sum_j \frac{1}{2}\hbar\omega_j\hat{p}_j^2+\sum_j \frac{1}{2}\hbar\omega_j\hat{q}_j^2, \tag{2a}$$

$$\hat{H}_f=\sum_j \frac{1}{2}\hbar\omega_j\hat{p}_j^2+\sum_j \frac{1}{2}\hbar\omega_j(\hat{q}_j-\Delta_j)^2+\hbar\omega_0. \tag{2b}$$

Here, the index j denotes each nuclear mode, with frequency ω_j, coordinate displacement between two states specified by Δ_j, and an energy splitting specified by $\hbar\omega_0$. Dimensionless p_j and q_j are used for convenience.

The dynamics of an electronic subsystem can be expressed in terms of a reduced density matrix $\boldsymbol{\rho}(t)$,[1,15] obtained by tracing over the nuclear coordinates. The off-diagonal terms, the electronic coherences, decay on a time scale that is determined after the nuclear trace. This time scale can be characterized via the nuclear overlap/phase function (NOPF) for an initial nuclear state evolving on two alternative PES's, given by

$$J(t)=\langle\Psi_i(t)|\Psi_f(t)\rangle \tag{3a}$$

$$=\langle\Psi_i(0)|\exp_+\left[-\frac{i}{\hbar}\int_0^t d\tau\,\Delta\hat{H}_{fi}(\tau)\right]|\Psi_i(0)\rangle. \tag{3b}$$

Here $\exp_+$ represents the time-ordered exponential, $\Psi_i(0)$ is an initial nuclear wave function, and $\Delta\hat{H}_{fi}(t)$ is

$$\Delta\hat{H}_{fi}(t)=\exp\left[\frac{i}{\hbar}\hat{H}_i t\right]\Delta\hat{H}_{fi}\exp\left[-\frac{i}{\hbar}\hat{H}_i t\right], \tag{4}$$

where $\Delta\hat{H}_{fi}=\hat{H}_f-\hat{H}_i$.

We invoke the Condon approximation between the electronic and nuclear parts in the reduced density matrix due to the difference of their time scales,[16] and consider a thermally equilibrated initial nuclear configuration. Using $\langle\cdots\rangle_T$ to represent thermal averaging over nuclear configurations, the thermally averaged form of $J(t)$ is given as

$$\langle J(t)\rangle_T=\left\langle\exp_+\left[-\frac{i}{\hbar}\int_0^t d\tau\,\Delta\hat{H}_{fi}(\tau)\right]\right\rangle_T. \tag{5}$$

If one next performs a cumulant expansion[17] of Eq. (5), and uses the fact that a second order truncation gives the exact result for the harmonic bath,[15] the result is

$$\langle J(t)\rangle_T=\exp\left[-\frac{it}{\hbar}\langle\Delta\hat{H}_{fi}\rangle_T-\frac{\langle(\delta\hat{U})^2\rangle_T}{\hbar^2}\int_0^t d\tau_2\int_0^{\tau_2}d\tau_1\,C(\tau_1)\right]. \tag{6}$$

In Eq. (6), the energy gap fluctuation autocorrelation function $C(t)$, which represents the response of the bath to a perturbation, has been introduced[3,18,19] and is defined as

$$C(t)=\frac{\langle\delta\hat{U}(t)\cdot\delta\hat{U}(0)\rangle_T}{\langle\delta\hat{U}(0)\cdot\delta\hat{U}(0)\rangle_T}, \tag{7}$$

where the fluctuation operator of electronic energy gap $\delta\hat{U}(t)$ is given as

$$\delta\hat{U}(t)=\Delta\hat{H}_{fi}(t)-\langle\Delta\hat{H}_{fi}\rangle_T. \tag{8}$$

We are interested in the decay of the norm of $\langle J(t)\rangle_T$, and the solution of Eq. (6) in the present model system can be found elsewhere.[9,20] The result for general bath spectral density $\Omega(\omega)$ is

$$|\langle J(t)\rangle_T|=\exp\left[-\int_0^{\infty}d\omega\,\Omega(\omega)\frac{\coth(\beta\hbar\omega/2)(1-\cos(\omega t))}{\omega^2}\right], \tag{9}$$

where $\beta=1/k_BT$ and

$$\Omega(\omega)=\sum_j \frac{1}{2}\Delta_j^2 w_j^2\,\delta(\omega-\omega_j). \tag{10}$$

The spectral density determines the coupling of bath degrees of freedom to the subsystem. Several forms for the spectral density have been proposed.[9] Here, we use the ohmic spectral density, for its analytic advantages, which is given as

$$\Omega(\omega)=\eta\omega\exp\left[-\frac{\omega}{\omega_c}\right], \tag{11}$$

where ω_c is a cutoff frequency and the amplitude η is a classically measurable friction coefficient of the bath, which reflects the amplitudes of the nuclear displacements.

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III. COMPARISON BETWEEN FQM AND MQC RESULT FOR CANONICAL NUCLEAR OVERLAP/PHASE FUNCTION

The incorporation of the ohmic spectral density into Eq. (9) leads to[21,22]

$$|\langle J^{qm}(t)\rangle_T| = \exp\left[-\eta\left\{ \frac{1}{2}\ln(1+(\omega_c t)^2) + \ln\prod_{n=1}^{\infty}\left[1+\frac{(\omega_c t)^2}{(1+n\beta\hbar\omega_c)^2}\right]\right\}\right] \tag{12a}$$

$$= \exp\left[-\eta\left\{ \frac{1}{2}\ln(1+\omega_c^2 t^2) + 2\ln\Gamma\left(1+\frac{1}{\beta\hbar\omega_c}\right) - 2\ln\left|\Gamma\left(1+\frac{1+i\omega_c t}{\beta\hbar\omega_c}\right)\right|\right\}\right], \tag{12b}$$

where superscript qm is used to distinguish the FQM solution from the MQC result. Although Eq. (12) is the exact solution, the various contributions discussed in the Introduction are not transparent. These can be identified by comparing the exact quantum solution and the MQC solution.

In the basic MQC methods, electronic states are described quantum mechanically, but nuclear degrees of freedom are treated classically. $\Delta\hat{H}_{fi}(t)$ is then a time-dependent function. In this case $|\langle J^{\mathrm{mqc}}(t)\rangle_T|$ is given by

$$\langle J^{\mathrm{mqc}}(t)\rangle_T = \left\langle \exp\left[-\frac{i}{\hbar}\int_0^t d\tau\, \Delta H_{fi}^{\mathrm{mqc}}(\tau)\right]\right\rangle_T, \tag{13}$$

where $\Delta H_{fi}^{\mathrm{mqc}}(t)$ is

$$\Delta H_{fi}^{\mathrm{mqc}}(t) = -\sum_j \hbar\omega_j\Delta_j\{q_j(0)\cos(\omega_j t) + p_j(0)\sin(\omega_j t)\} + \sum_j \frac{1}{2}\hbar\omega_j\Delta_j^2 + \hbar\omega_0. \tag{14}$$

The result is readily obtained from direct calculation with a classical phase space weight, or from a high temperature limit of the expressions above, leading to

$$|\langle J^{\mathrm{mqc}}(t)\rangle_T| = \exp\left[-\eta\left\{ \frac{2t}{\beta\hbar}\tan^{-1}(\omega_c t) - \frac{1}{\beta\hbar\omega_c}\ln(1+(\omega_c t)^2)\right\}\right]. \tag{15}$$

Since there is no nuclear quantum effect in the MQC method, the decay of $|\langle J^{\mathrm{mqc}}(t)\rangle_T|$ is only from the dephasing associated with a classical thermal distribution of electronic energy gaps. To obtain a characteristic classical dephasing time (cdp), we consider a short-time/strong coupling case.[13] Then $|\langle J^{\mathrm{mqc}}(t)\rangle_T|$ is approximated as

$$|\langle J^{\mathrm{mqc}}(t)\rangle_T| \approx \exp\left[-\frac{t^2}{2\tau_{\mathrm{cdp}}^2}\right], \tag{16}$$

where

$$\tau_{\mathrm{cdp}} = \sqrt{\frac{\beta\hbar}{2\eta\omega_c}}. \tag{17}$$

The same expression can be obtained directly from the short-time expansion of the exponent in $J(t)$, Eq. (9), by evaluating only the leading term in a high temperature expansion for the present model.

In Eq. (17) we can see that this dephasing time is proportional to $T^{-1/2}$; i.e., the decay time decreases with increasing temperature. The decay time is also simply related to the energy gap fluctuation. From linear response theory, the energy gap fluctuation is

$$\langle \delta\hat{U}^2\rangle_T = \frac{2\lambda}{\beta}, \tag{18}$$

where the reorganization energy λ for the ohmic bath is[23]

$$\lambda = \eta\hbar\omega_c. \tag{19}$$

Thus, we find in the present case

$$\tau_{\mathrm{cdp}} = \frac{\hbar}{\sqrt{\langle \delta\hat{U}^2\rangle_T}}. \tag{20}$$

Equation (20) clearly shows that the electronic dephasing at short time is associated only with the thermal distribution of energy gaps; the temperature dependence is just that of the width of the distribution. The solution in the short-time limit does not reflect any dynamics of ΔH_{fi}^{cl}.

In the context of MQC dynamics implementations, it is important to emphasize that the decay displayed above ensues from the thermal average over only the initial value of the phase in the exponent of Eq. (13). It is only when one averages over a group of trajectories with different initial phases, to calculate the thermally averaged NOPF, that the differences among phase factors for different trajectories cause this dephasing. Typical MQC implementations carry out individual trajectories from individual phase points, averaging results over initial conditions based on the set of trajectories only at the end. Hence, the dephasing contribution reflected in Eq. (17) should not be taken into account in these individual trajectories.

Now we consider corrections to the MQC approach. Considering only the first correction for small $\beta\hbar\omega$, one easily finds

$$|\langle J^{qm}(t)\rangle_T| = \exp\left[-\eta\left\{ \frac{2t}{\beta\hbar}\tan^{-1}(\omega_c t) - \frac{1}{\beta\hbar\omega_c}\ln(1+(\omega_c t)^2) + \frac{1}{6}\beta\hbar\omega_c^3 t^2 + \vartheta(\beta^3, t^2)\right\}\right]. \tag{21}$$

We introduce here a "decoherence function" $D(t)$.[4] The function is defined by the relation

$$J^{qm}(t) = J^{\mathrm{mqc}}(t)D(t). \tag{22}$$

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In the present case, the thermally averaged $\langle J^{qm}(t)\rangle_T$ is then expressed for small β with

$$|\langle D(t)\rangle_T| = \exp[-\tfrac{1}{6}\beta\eta\hbar\omega_c^3 t^2 + \vartheta(\beta^3, t^2)]. \tag{23}$$

Neglecting third and higher orders in β, the electronic decoherence function then becomes

$$|\langle D(t)\rangle_T| \approx \exp\left[-\frac{t^2}{2\tau_{\mathrm{qdc}}^2}\right], \tag{24}$$

where τ_{qdc} is a decoherence time[3] corresponding to the short time/high temperature contribution from nuclear quantum effects. It is given by

$$\tau_{\mathrm{qdc}} = \sqrt{\frac{3}{\beta\eta\hbar\omega_c^3}}. \tag{25}$$

Since this result is the leading correction in β to the MQC result, it can also be evaluated for this model directly from the short-time expansion of the exponent in $J(t)$, Eq. (9), from the second term in a high temperature expansion. The first such term yields the MQC result, as already noted, so the second term yields the leading correction. Equation (25) shows that as the temperature increases, this electronic decoherence time scale becomes longer and the electronic decoherence becomes weaker. At low temperature, of course, the higher order terms in β would need to be considered.

From Eq. (17) and Eq. (25), we can obtain a relationship between this decoherence time and the classical bath dephasing time. Comparing τ_{qdc} and τ_{cdp} leads to

$$\frac{\tau_{\mathrm{qdc}}}{\tau_{\mathrm{cdp}}} = \frac{\sqrt{6}}{\hbar\omega_c\beta}. \tag{26}$$

The ratio is proportional to the ratio of two time scales $1/\omega_c$ and $\hbar\beta$. Reichman and Silbey[20] called the former time scale the ''mechanical'' time scale and the latter the ''quantum-thermal'' time scale. $1/\omega_c$ is a measure of the fastest natural time scales in the bath, corresponding to the modes with the largest quantum effect. The ratio expresses the fact that at low temperature, the classical bath dephasing contribution becomes small compared to the contribution associated with bath quantum effects, and conversely at high temperature. At room temperature, the quantum-thermal time scale $\hbar\beta$ is approximately 25 fs. Nevertheless, in real solutions, there are many examples of high frequency nuclear motions, particularly in polar condensed phase systems. For example, OH stretching vibrations of water molecules cause high frequency fluctuations in the dipole moment on a time scale of about 10 fs. Their involvement in the electronic relaxation of a hydrated electron produces significant quantum contributions.[7,8,24]

One other relation among time scales is of interest. The real part of $C(t)$ in Eq. (7) is related to solvation dynamics, and the experimental fluorescence Stokes shift function.[25] In the high temperature case, where $\beta\hbar\omega_c \ll 1$, it is approximated in the ohmic bath as[26]

$$\mathrm{Re}\{C(t)\} \approx \frac{1}{1+\omega_c^2 t^2}. \tag{27}$$

Stratt and Cho[19] showed, in the same classical regime, that $\mathrm{Re}\{C(t)\}$ can generally be written as a Gaussian form in the short time, inertial, limit, here $\omega_c t \ll 1$,

$$\mathrm{Re}\{C(t)\} \approx \frac{1}{1+\omega_c^2 t^2} = 1 - \omega_c^2 t^2 + \cdots,$$
$$\approx \exp[-\omega_c^2 t^2] = \exp\left[-\frac{t^2}{2\tau_g}\right], \tag{28}$$

where the Gaussian solvation time scale τ_g is given as

$$\tau_g = \frac{1}{\sqrt{2}\omega_c}. \tag{29}$$

This result is simply that which could be obtained directly from the leading term in the high temperature/short time limit of $\mathrm{Re}\{C(t)\}$, Eq. (7), evaluated for the present model.

Comparison between τ_{qdc} and τ_g leads to

$$\left(\frac{\tau_{\mathrm{qdc}}}{\tau_g}\right)^2 = \frac{6}{\beta\eta\hbar\omega_c} = \frac{6}{\beta\lambda}, \tag{30}$$

Prezhdo and Rossky[3] have arrived at the same high temperature, short-time relationship between τ_{qdc} and τ_g, more generally, using the frozen Gaussian approximation.[13,27,28]

IV. DERIVATION OF DECOHERENCE TIME SCALES USING THE WIGNER DISTRIBUTION

In this section, we investigate, for the spin-boson model with the ohmic spectral density, how a decoherence time scale associated with quantum bath contributions arises when the quantum treatment is implemented via a combination of the canonical Wigner distribution and MQC methods.

The canonical Wigner distribution function is well known for the harmonic oscillator model.[15] When we use Eq. (2a) as the initial Hamiltonian, the Wigner distribution function is given as

$$\rho_W(\mathbf{p},\mathbf{q}) = \prod_j \frac{\tanh(\beta\hbar\omega_j/2)}{\pi} \exp[-\tanh(\beta\hbar\omega_j/2)(p_j^2+q_j^2)]. \tag{31}$$

Applying the Wigner distribution to the MQC NOPF of Eq. (13) and incorporating the ohmic spectral density into the equation leads to

$$|\langle J^{\mathrm{mqc}}(t)\rangle_T^W| = \exp\left[-\eta\left\{\frac{1}{2}\ln(1+\omega_c^2 t^2) + 2\ln\Gamma\left(1+\frac{1}{\beta\hbar\omega_c}\right) - 2\ln\left|\Gamma\left(1+\frac{1+i\omega_c t}{\beta\hbar\omega_c}\right)\right|\right\}\right] \tag{32a}$$

$$= \exp\left[-\eta\left\{\frac{2t}{\beta\hbar}\tan^{-1}(\omega_c t) - \frac{1}{\beta\hbar\omega_c}\ln(1+(\omega_c t)^2) + \frac{1}{6}\beta\hbar\omega_c^3 t^2 + \vartheta(\beta^3, t^2)\right\}\right], \tag{32b}$$

where $\langle\cdots\rangle_T^W$ represents thermal averaging over nuclear configurations weighted by the Wigner distribution function.

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The exponent of Eq. (32b) reveals the identical decoherence term as in Eq. (21), as pointed out by Lockwood *et al.*[10] More notable is that the result in Eq. (32a) is identical to the exact FQM result of Eq. (12b). Thus the magnitude of the decoherence function [Eq. (22)] is precisely given by this approach. This could be expected based on the earlier results of Egorov *et al.*,[29] where it was shown that the Wigner distribution combined with classical propagation provides the exact rate of electronic state-to-state transitions for this model, for constant off-diagonal electronic coupling. However, it is interesting to note how the Wigner distribution function introduces quantum corrections to the classical result.

Comparing the Wigner distribution function $\rho_W(\mathbf{p},\mathbf{q})$ with the classical result shows that the Wigner distribution is just the classical distribution convoluted with a Gaussian of the form

$$\exp[-(p^2+q^2)/2\sigma^2], \tag{33}$$

having a characteristic squared width given by

$$\sigma^2 = \tfrac{1}{2}[\coth(\beta\hbar\omega/2) - (2/\beta\hbar\omega)]. \tag{34}$$

That is, the contributions from quantum fluctuations associated with the bath are exactly included in the NOPF for the model if each individual initial classical phase point is broadened by the width σ.

It is exactly this spatial wave packet width of Eq. (34) that was introduced in the semiclassical frozen Gaussian trajectory approach to the NOPF by Neria and Nitzan.[13] They based the choice on the analytical result in the strong coupling/high temperature case for the displaced harmonic oscillator model. The physical significance of that choice is clear here. The fact that the choice does not give the exact result in general for the model[13] is also clear here, since the momentum distribution used for wave packet propagation in the earlier approach is classical. The same width was adopted by Prezhdo and Rossky[4] to introduce quantum bath corrections to MQC nonadiabatic transition rates, in the form of electronic coherence dissipation. In both cases, the errors associated with the method are small at high temperature, since the calculations are carried out for the case that the initial momentum on initial and final electronic states is the same; then, the momentum does not explicitly contribute to the decay rate of the NOPF at short times.[4]

It is noteworthy that the same width expression arises naturally in developing a local harmonic approximation for quantum fluctuations. The Feynman and Kleinert variational perturbation theory[30] is based on introducing an optimal local harmonic description of quantum fluctuations around path integral centroid positions. The formulation introduces exactly the same functional form for σ, a smearing width, with the (local) frequency determined variationally. For a system that is, in fact, harmonic over a distance characterized by this width, the width would be given directly by the expression for σ given in Eq. (34).

Finally, it should be noted that although the magnitude of the MQC NOPF associated with the Wigner distribution function in Eq. (32b) is exactly the same as the FQM result in Eq. (12), $\langle J^{\mathrm{mqc}}(t)\rangle_T^W$ differs from $\langle J^{qm}(t)\rangle_T$ even for the present model, since the phase factor of $\langle J^{\mathrm{mqc}}(t)\rangle_T^W$ and $\langle J^{qm}(t)\rangle_T$ are different.[29]

V. SUMMARY

In this study, the spin-boson model with the ohmic spectral density has been used to analyze electronic dephasing with particular attention to the limitations of mixed quantum-classical simulation. For the short-time/strong coupling case, we showed a relationship between an electronic decoherence time attributed to quantum bath contributions and that due to electronic dephasing for a classical bath. The relationship leads to two time scales, also considered earlier,[20] and provides a simple criterion for determining the importance of accounting for quantum bath effects as a correction to a trajectory-based MQC method.

With the same model, we also showed that the relationship between the electronic dephasing time and a solvation dynamics time scale has the same form as one proposed based on frozen Gaussian dynamics in a previous study.[3]

Finally, we have also explicitly shown that an MQC method associated with a Wigner distribution of initial conditions manifests all of the contributions to electronic decoherence for the model. The quantum bath contribution to electronic decoherence can be exactly accounted for with a distribution of quantum fluctuations around each classical phase space point, associated with quantum mechanical uncertainty. The spatial distribution of quantum fluctuations that is evident here is the same as that appearing in the Feynman–Kleinert variational local harmonic description,[30] and also implemented in approximate trajectory-based semiclassical treatments of coherence dissipation.[4,7,8]

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